

# Chia-Hung Kuo

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## Data Scientist | Causal Inference & Experimentation

PhD Candidate in Economics from Michigan State University with experience in A/B testing, uplift modeling, and causal inference. Currently building production statistical systems for an NGO in Taiwan where analysis directly drives annual government funding allocations to universities.

## Experience

### Taiwan Assessment and Evaluation Association, Taipei, Taiwan

Data Science Consultant

09/2024-Current

- Modernized data infrastructure from legacy FTP servers to GCP (GCS, BigQuery) using Python, SQL, and dbt; reduced full analysis cycle from ad hoc to one day
- Built two anomaly detection systems informing government funding decisions across 1,600+ annual evaluations: a time-series regression flagging irregular department scoring patterns, and a latent variable model separating genuine quality from reviewer-specific bias; delivered via Tableau to non-technical government stakeholders

### Michigan State University, East Lansing, Michigan

Lecturer & Teaching Assistant

09/2018-06/2024

- Taught and assisted microeconomics, macroeconomics, international finance courses ranging from 300+ student lectures to 30-person seminars
- Excellence in Teaching Award (2022, awarded to one PhD student annually across the department); Honorable Mention, Peter Schmidt Award (2019; ranked 2nd in PhD cohort qualifying exams)

## Projects & Research Papers

### A/B Testing with Uplift Modeling and Causal Decomposition | [Live App](#)

11/2025

- Implemented S/T/X/R meta-learners using EconML and XGBoost on 13.9M-user RCT; X-Learner best performer (Qini: 0.1753), top decile converting at 5x control rate
- Decomposed CATE via IV into reachability  $P(E) \times$  responsiveness  $LATE(X)$ , enabling differentiated bidding strategy; quantified 10% ranking cost for strategic optimization

### Multi-Target Stacking: A Novel Ensemble Framework

09/2025

- Developed novel ensemble framework exploiting auxiliary target diversity across 14M+ financial time series records using XGBoost and scikit-learn base learners; 57% improvement and 47% variance reduction over standard stacking
- Validated via 84-fold walk-forward CV with Optuna hyperparameter tuning and MLflow experiment tracking; maintained stable 95.6% PR-AUC ( $\pm 6.8\%$ ) on held-out 2024 test year despite distribution shifts

## Research Papers

2021-2024

- Applied DiD-IV strategy exploiting China's 1986 Compulsory Schooling Law (25 provinces, 16,500+ individuals); estimated 0.84 SD increase in internal locus of control from completing junior high school
- Estimated structural discrete choice model (MLE) on NLSY97 (track ~9,000 individuals for 20 years); identified 12% reduction in migration for two-parent households, driven by 35% higher home premium

## Education

PhD Candidate (ABD) in Economics, Michigan State University, East Lansing, MI

06/2024

MSc in Economics, University of Bonn, Bonn, Germany

06/2018

BSc in Quantitative Finance, National Tsing Hua University, Hsinchu, Taiwan

06/2014

## Technical Skills

Python, SQL, Git, pandas, PySpark, scikit-learn, XGBoost, EconML, GCP, AWS, dbt, MLflow, Docker